Transport Priority and Composition Effects in Liver Transplantation

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Institutional Setting

Institutional setting – Green Path

- China's Green Path policy granted donor organs priority transport across air, rail, and road channels.
- Objective: reduce transport delays and therefore cold ischemia time (CIT).
- Relevance: liver graft viability is highly sensitive to CIT. True for other organs too

CIT = packaging at source hospital
+ handoffs
+ ground transport
+ air/rail
+ Ground Transport
s+ processing at receiving hospital

Institutional Background

- Hospitals with an organ check if they have a patient who needs it
  - Patient must meet minimum "medical urgency" threshold (m)
- If not, check local hospitals
- If not, then ship organ to another region that has a qualifying patient
- Demand always exceeds supply. However, organs may be wasted because of long transport duration
- Green Path can reduce transportation time and provide value

Theory

Motivation: Empirical puzzle

The raw data generate a contradiction:

- nonlocal organs experience lower transport time after Green Path;
- system-wide mean CIT rises;
- conditional survival declines.

Contribution:

Show that these are not inconsistent by separating:
intensive margin
from
composition and equilibrium re-sorting.

Theory: Consumer / Patient Model

Two hospitals $j \in \{1, 2\}$, with $s_1 > s_2$ and viability thresholds $q_1 < q_2$.

$$q^{\mathrm{del}} = q - \delta I, \qquad q \in \{q_h, q_l\}, \qquad \delta > 0.$$

Patients differ in:

- urgency $m \sim U[0, 1]$,
- quality sensitivity $\theta \sim U[0, 1]$.

Patient utility:

$$U_{ij} = \theta\big(s_j + \mathbb{E}[q_j^{\mathrm{del}}]\big) - p_j - \mathbf{1}(i \neq j)c.$$

Hospitals choose prices p_j strategically, anticipating demand and the organ supply they can actually use.

Composition Effects: Intuition

k is the set of organs that become available for transplant at a hospital.

- Average realized quality:

$$\mathbb{E}[q^{\text{del}}] = \sum_k w_k q_k^{\text{del}},$$

  with type weights w_k.
- Change in average quality can be decomposed as:
- within type improvement + composition shift
- The paper's logic is that the first term is positive, but the second can be negative and larger in magnitude.
- Expected value ignores market expansion, i.e. more people are being served

Theory Results

- P1-3: *What's the impact of minimum urgency thresholds?* Prices are lower with higher minimum urgency thresholds.
  - Firms do not have incentive to set any higher requirements.
  - Medical criteria can act to impact market power / competition
- P4: Green Path can increase or decrease hospital profits. Why?
  - Role of quality and supply: Supply $\Uparrow$, but average quality of (accepted) organs $\Downarrow$
  - Key issue: Cannot (legally) condition prices on realized quality, only on *expected quality*
  - So if distribution of quality $\Downarrow \implies$ patient WTP for organ $\Downarrow \implies \pi \Downarrow$
  - Even though organ supply has only expanded, so all original ones are still there
  - So profits can decrease even without competitive effects
- P5: Green Path can increase CIT (or reduce average transplanted quality)
- (suggestion) P6: Market expansion – who in model would have received organs who does not receive them now?

Green Path effect: Intuition

In equilibrium, Green Path can raise average CIT when

$$\frac{\text{nonlocal share after}}{\text{nonlocal share before}} > \frac{CIT(nonlocal)_{\text{before}} - CIT(local)_{\text{before}}}{CIT(nonlocal)_{\text{after}} - CIT(local)_{\text{after}}}.$$

Interpretation:

- **direct transport effect:** each nonlocal organ arrives faster;
- **composition effect:** many more nonlocal and marginal organ are now feasible;
- **aggregate outcome:** the weight shift can dominate the per-organ gain.

Empirical

RDiT identification

The paper estimates a regression discontinuity in time around the national implementation date:

$$Y_t = \alpha + \delta_1 GP + \delta_2 nonlocal + +\delta_3 GP \times nonlocal + ... + \varepsilon_t.$$

Key identifying claims:

- no precise manipulation of transplant timing at the cutoff;
- balance / continuity in observed donor and recipient covariates;
- falsification checks using alternative cutoffs.

Appendix reports continuity tests and density checks consistent with the design.

Latent provenance problem

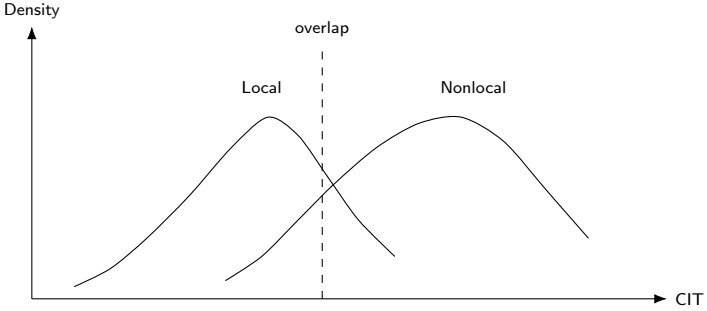
- **Empirical challenge: donor location is not directly observed**
- Solution is a mixture model on the CIT distribution:

$$f(\mathrm{CIT}) = \pi_L f_L(\mathrm{CIT}) + \pi_N f_N(\mathrm{CIT}), \qquad \pi_L + \pi_N = 1.$$

Intuition:

- local and nonlocal organs generate different CIT modes;
- Green Path shifts the nonlocal component;
- classification uses distributional shape rather than provenance labels.

EM Intuition: Overlapping CIT Distributions



Mixture: $f = \pi_L f_L + \pi_N f_N$
E-step: assign probabilities
M-step: update densities + shares

Each observation gets a probability of being nonlocal based on how likely it is under each distribution

Regression: Does CIT reduce or increase with Green Path?

DV: CIT	(1)	(2)	(3)	(4)
Time Trend	Linear	Quadratic	Linear	Quadratic
Sample	All	All	Center1	Center1
Month Bandwidth	[-43,43]	[-43,43]	[-43,43]	[-43,43]
Panel A: All samples (simple classification of nonlocal based on closeness to group center)				
greenpath	-1.391	-3.777	2.074	-2.613
	(4.789)	(5.016)	(5.241)	(5.602)
nonlocal	171.937***	173.675***	198.469***	203.223***
	(5.497)	(5.675)	(7.178)	(7.652)
greenpath*nonlocal	-14.058**	-17.335***	-30.479***	-36.027***
	(5.894)	(6.245)	(8.193)	(8.675)
Panel B: Removed samples that are most likely misidentified				
greenpath	8.929	-1.009	3.792	-2.266
	(5.559)	(5.689)	(4.478)	(6.439)
nonlocal	273.037***	276.431***	291.871***	293.758***
	(4.719)	(4.703)	(6.598)	(6.424)
greenpath*nonlocal	-37.954***	-37.488***	-60.327***	-59.532***
	(5.510)	(5.583)	(6.044)	(6.471)
Recipient characteristics	Yes	Yes	Yes	Yes
Donor characteristics	Yes	Yes	Yes	Yes

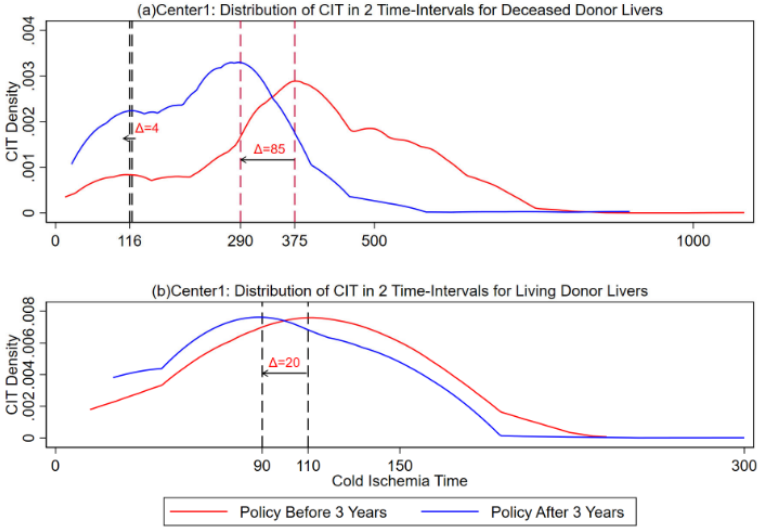
Main results: what lines up with the theory

- Nonlocal transport time falls substantially; savings of 40–60 minutes.
- Nonlocal share and transplant volume increase
- Marginal organ usage rises, especially at the higher-quality hospital.
- Average realized CIT rises.
- Recipient-conditional survival worsens.

Big Picture

Empirical observations broadly consistent with theory, which helps us understand the "why" for these data patterns

More Evidence



Thoughts and Suggestions

Things to Like

- Paper studies a very important question at intersection of healthcare policy, firm responses, consumer choices
- Seeming empirical puzzle resolved by identifying the mechanism and disentangling effects
- Lots of care in both model development and in the empirical section (more in appendix)
- Model accounts for most first order aspects of problem
- Nice use to EM to recover distributions since donor location unknown
- Regression Discontinuity makes sense here
- Overall: Helps understand what we should expect when policies have complex responses by multiple economic agents

In Sum:

Lots of things to like about the paper!

Thoughts – Theory

- Theory model is stylized with some aspects that would be good to think about
  - Organ quality and hospital quality are substitutes from consumer's view. True from a medical viewpoint.
  - How to model market expansion?
  - How should medical condition play a role? Currently, it is a bit passive (extension: correlation between m and valuation). What is the outside option while waiting?
  - What aspects depend on competition? Most results (directionally) apply even to monopolist?
  - Is anybody worse off in expectation?
  - Is there a way to characterize how the supply / demand imbalance impacts these outcomes? Comparative statics

Thoughts – Empirical and General

- Empirics looks only at two hospitals.
- Outcome is conditional on an endogenous selection margin, so must be interpreted with care – who else would have entered market after?
- Identify "equilibrium effects" of policy, not "treatment effects" (maybe a good thing)
- What additional things did hospitals do after this policy?
- RDIT reasonable with sharp timing at a national level
- Seems like a reasonable place to check robustness using Difference-in-Differences (nonlocal versus local)
  - Can get direct transport effect (use Post $\times$ Nonlocal probability)
  - However, there are spillovers between local and nonlocal groups
  - Maybe long-distance and short-distance organs
- Should government keep minimum medical threshold the same with Green Path?

Overall

Empirics connects very nicely with Theory. Would be great to see more work like this that thoughtfully create complementarities between theory and empirics.